

The Political Economy of State Economic Development: A Path to Unproductive Entrepreneurship^{*}

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Abstract

Entrepreneurship is a key driver of economic growth, and governments actively promote policies aimed at fostering it. However, these policies may also incentivize rent seeking by shifting relative economic returns from what is productive to what is not, accomplishing the very opposite of the stated intention. Leveraging 40,000 geocoded establishments in the lobbying industry from the National Establishment Time Series database, we explore how unproductive economic activity (proxied by lobbying-industry activity) changes in states after they award their first “extraordinarily large” business incentive. We find that, relative to states that have not awarded such incentives, establishment-level lobbying-industry employment increases by 3.6%, with larger effects in capital counties where lobbyist employment concentration expands significantly. In contrast to other studies that fail to find significant employment gains arising from economic development incentives, we do find them—but in lobbying and unproductive entrepreneurship.

Keywords: Economic Development Incentives, Rent Seeking, Unproductive Entrepreneurship, Lobbying

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1 Introduction

Entrepreneurship is a key driver of economic growth. As a result, governments now actively adopt policies with the stated intention of fostering it. However, many of these policies not only fail to deliver on the stated intention, they actually lead to reductions in productive economic activities and, indeed, foster crony capitalism (Audretsch and Fiedler, 2023; Cumming and MacIntosh, 2006; Pierrakis and Saridakis, 2017; Boudreaux, Nikolaev and Holcombe, 2018; Gustafsson, Tingvall and Halvarsson, 2020; Sobel, 2008; Holcombe, 2013; Klein et al., 2022). Government involvement in business development has led scholars to call for greater consideration of the public choice aspects of related policies (Karlson, Sandström and Wennberg, 2021; Lucas and Boudreaux, 2020; Kärnä et al., 2023). In response to that call, our research is the first to examine the public choice aspects of large economic development incentives, specifically their role in fostering unproductive entrepreneurship. By doing so, we hope to shed light on the relationship between public choice considerations and economic policies, in particular those that fail to accomplish their stated purpose.

Various economic development incentives are used by U.S. states to encourage private-sector firms to locate, expand, or stay within their geographic borders. The aggregate value of these incentives has tripled since 1990 and now exceeds more than \$40 billion annually (Bartik, 2019). Incentives valued in the billions awarded to individual firms have also become increasingly common, with New York’s \$6.4 billion incentive to Micron Technology in 2022 and Wisconsin’s \$4.8 billion incentive to Foxconn in 2017 as examples that attracted national attention. According to the Good Jobs First Subsidy Tracker database, at least twenty subsidies valued at one billion dollars or more have been awarded to individual companies since 2010.¹

Elected officials, recipient firms, and government agencies often justify large public subsidies based on claims of additional jobs, higher incomes, and expanded tax revenues (Poole et al., 1999). Despite these contentions, there is scant evidence that the subsidies yield these benefits, leading many to question subsidy programs’ efficiency, transparency, and accountability (Hicks and Shughart, 2007; Fox and Murray, 2004; Gabe and Kraybill, 2002; Peters and Fisher, 2004; Bingham and Bowen, 1994; Bundrick and Snyder, 2018; Mitchell, Sutter and Eastman, 2018; Bartik, 2018; Bartik, 2019). To the extent that subsidies do result in gains, the empirical evidence suggests

¹The subsidy tracker database is available at <https://subsidytracker.goodjobsfirst.org> [accessed: 3/20/2023].

the gains are captured by recipient firms and their shareholders, and, in the form of increased campaign contributions and electoral support, by the incumbent politicians who award them (Sobel, Wagner and Calcagno, 2022).

As Baumol (1990) and Tullock (1967) both argue, there are costs associated with these incentives, such as deadweight losses from rent seeking (to secure subsidies), corruption, and, potentially, the reallocation of productive talent. Despite their wide acknowledgment, these costs are difficult to quantify and are therefore frequently omitted from policy discussions. A full accounting of the benefits and costs of incentives is likely to show that these programs are a net negative for society as a whole (Coyne and Moberg, 2014; Hicks and Shughart, 2007; Mitchell, Sutter and Eastman, 2018; Felix and Hines, 2013; Sobel, 2008).

In the case of entrepreneurial effort, for instance, Baumol (1990) theorizes that entrepreneurs allocate their time and talents between “productive activities such as innovation and largely unproductive activities such as rent seeking or organized crime,” based in part on the relative payoffs of the activities. Stated differently, if an entrepreneur believes the highest and best use of their labor is to pursue economic development incentives rather than to innovate privately, they may rationally decide to reallocate their efforts to unproductive activities – such as lobbying and rent seeking – to increase the likelihood of being awarded a subsidy.

Building on the recent work of Sobel, Wagner and Calcagno (2022) and Dove (2021), who study how incentive-giving affects the political landscape at the state level, we extend the literature on the political costs of economic development incentives in several dimensions. First, rather than focusing on more aggregate economic outcomes, we leverage a national geocoded data set of establishments, the National Establishment Time Series (NETS), and explore how state incentive-giving behavior affects individual establishments within the lobbying industry. Our full sample includes more than 40,000 establishments across all 50 states from 1997 to 2019, providing us with a comprehensive micro-perspective.

Second, we provide a more nuanced view than does state-level analysis because we know the location of individual establishments. If development incentives do, in fact, encourage unproductive entrepreneurship via rent seeking, we expect to observe more clustering in lobbying activity in and around state capitals even when the state’s overall level of lobbying activity remains unchanged.

Specifically, we utilize a difference-in-differences identification strategy that adjusts for staggered

timing across states to investigate how a state’s first “extraordinarily large” incentive affects the growth and spatial organization of the state’s lobbying industry. We follow Sobel, Wagner and Calcagno (2022) and define an “extraordinarily large” incentive, outlined in detail in Section 3, to be the first instance in which a state awards an incentive to a single private-sector firm that is *substantially* larger than the state’s own past giving behavior.² States that have not (to date) awarded an “extraordinarily large” incentive are used as counterfactual comparisons.

We document two important results that are consistent with the assertion that economic development incentives trigger more unproductive activity. First, relative to states that have never awarded an extraordinarily large economic development incentive, establishment-level lobbyist employment increases an average of 3.6% after a state awards its first extraordinarily large incentive. Existing lobbyist establishments located in capital counties appear to be the main driver of this effect, experiencing establishment-level gains of almost 5% (relative to employment at lobbying establishments located in capital counties of comparison states that do not offer these extraordinarily large incentives). Additionally, we find that lobbyist employment as a share of total private-sector employment increases in capital city counties by almost 6%. There is no evidence that large economic development incentives spur new lobbying establishments, implying that existing establishments in the state – primarily located in the capital county – “capture” the unproductive gains.

The following sections of the paper review previous work, outline the data and identification strategy, present the results, and offer concluding remarks.

2 Literature Review

The literature on economic development incentives widely concludes these incentives are, at best, weakly effective and, at worst, economically inefficient and distortionary due to the high opportunity cost. Academic research on these incentives has repeatedly questioned the effectiveness, efficiency, transparency, and accountability of these incentive programs, finding that they generally do not lead to significant changes in economic growth, unemployment, tax revenue, or personal income

²If an individual incentive is valued at \$50 million or more, then the Good Jobs First Subsidy Tracker defines this as a “megadeal incentive.” The problem with this definition is that a single \$50 million incentive is a much larger share of economic activity in Rhode Island than it is in Texas, which is why we prefer the term “extraordinarily large.” Our definition, described in Section 3, adjusts for size differences across states to normalize the value of individual subsidy awards in order to make them more directly comparable from state-to-state.

(Hicks and Shughart, 2007; Fox and Murray, 2004; Gabe and Kraybill, 2002; Peters and Fisher, 2004; Bingham and Bowen, 1994; Bundrick and Snyder, 2018; Mitchell, Sutter and Eastman, 2018; Bartik, 2018; Bartik, 2019). Perhaps tellingly, is that the state agencies that offer these incentives rarely perform follow-up assessments of the programs, which often leads to newspaper articles and public demands for audits and more transparency (Hinkley et al., 2000; Pew Charitable Trusts, 2016; Emont and Sebastian, 2023). Thus, distortionary secondary effects, increased rent-seeking, and unproductive entrepreneurship can result.³

The basic hypothesis we test in this paper stems from the public choice literature on rent seeking that originates in the seminal papers of Tullock (1967), Krueger (1974), and Posner (1975). These studies show that previous research underestimated the real cost to society of government taxes, regulations, and subsidies because it ignored how much money and resources organizations spend trying to influence political decisions about these policies through lobbying and other activities.

One method scholars have used to measure the extent of rent seeking was to compare economic activity in state capital counties to other similar counties (Mixon, 1995; Mixon Jr, Laband and Ekelund Jr, 1994; Sollars, 1996; Sobel and Garrett, 2002). We also employ that line of estimation. A subset of the literature on government favor-seeking, most notably Weitzel et al. (2010), Sobel (2008), Murphy, Shleifer and Vishny (1991), and Baumol (1990), compares the allocation of individual talent or effort between entrepreneurial activities, including productive and unproductive activities. This literature recognizes that individuals can choose whether to devote their time and effort to productive, positive-sum activities or to unproductive, zero- or negative-sum activities and, further, that they will choose the one that generates the highest personal return. This choice is driven in part by institutional structure such that whenever economic and political institutions make it easier for individuals to secure personal returns through government transfers, they will do so.

³For a more recent review of the literature on incentives for economic development, see Sobel, Wagner and Calcagno (2022). An issue to consider is whether any employment increases from firms that receive these large incentives can offset increases in employment from unproductive entrepreneurship that emerge. Although this may be an interesting cost-benefit analysis, the literature suggests that firms receive many times the value of the jobs they create. For instance, the cost per job for FedEx to create a new hub in North Carolina was \$77,000 per job (Calcagno and Hefner, 2018), Mercedes in Alabama was \$192,730 (Calcagno and Hefner, 2009), and \$170,000 for Cabela’s incentive in West Virginia (Hicks and Shughart, 2007). In Michigan, Hicks and LaFaive (2011) find a cost of \$123,000 per job created in construction, with 75% of these jobs lasting a year or less. In addition, Gabe and Kraybill (2002) note that, at the firm level, establishments tend to overstate their potential employment in an effort to secure these incentives.

Where institutions provide secure property rights, the rule of law, contract enforcement, and limits on governmental ability to transfer wealth through taxation and regulation, creative individuals will be more likely to engage in productive entrepreneurship activities that create wealth for both themselves and others (Sobel, 2008; Kreft and Sobel, 2005; Sobel, Clark and Lee, 2007; Coyne, Sobel and Dove, 2010). Stansel and Tuszynski (2018), Hall and Lawson (2014), and Lawson, Murphy and Powell (2020) all show these institutional guarantees are consistent with the concept of economic freedom.⁴ In contrast, in areas with lower economic freedom, these same individuals are instead more likely to engage in attempts to manipulate the political or legal process to capture transfers of existing wealth through unproductive political and legal entrepreneurship activities that destroy wealth (e.g. lobbying and lawsuits).⁵

To be clear, the reallocation occurs because institutional structure largely determines the relative personal and financial rewards of investing entrepreneurial energies into productive market activities versus investing them into unproductive (e.g., political and legal) activities.⁶

While Dove (2021) is the only paper to attempt to demonstrate empirically a correlation between state economic development incentives and unproductive entrepreneurial activity, there is some other closely related literature. For example, Tuszynski and Stansel (2018) find that targeted economic development incentives reduce overall entrepreneurial activity. Other papers such as Qian and Acs (2023) argue economic development policy should shift away from attempting to lead entrepreneurial activity and toward supporting activity in existing entrepreneurial ecosystems. Davidson and Ekelund Jr (1994) and Lucas and Fuller (2017) argue that the context of the institutional environment matters in determining whether entrepreneurial activity is productive, unproductive, or even destructive. Trubnikov (2021) argues for a slightly different typology from Baumol (1990) where unproductive and destructive are separated to note that rent seeking activity is always destructive and at best unproductive.⁷

⁴There is now a large literature demonstrating that policies consistent with economic freedom are important to expand the rate of productive entrepreneurship see (Sobel, 2008; Bennett and Long, 2019; Boudreaux and Nikolaev, 2019; Boudreaux, Jha and Escaleras, 2023; Bennett and Nikolaev, 2021; Bosma et al., 2018)

⁵See Minniti (2008) for a review of literature that examines the role of government policy on entrepreneurial activity.

⁶For instance, high levels of government corruption result in individuals reallocating effort toward these unproductive activities (Dutta and Sobel, 2016; Boudreaux, Nikolaev and Holcombe, 2018)

⁷Trubnikov (2021) argues that rent seeking might only be unproductive if entrepreneurs find ways to circumvent regulatory or other barriers to add value in that environment. More generally, the literature views rent-seeking and lobbying as welfare reducing because the policies that emerge are often targeted to specific firms or industries so they distort the economic calculus of consumers and firms. However, it is possible that lobbying could lead to broad-based

Our paper is among the first to attempt to quantify how the amount of rent seeking or unproductive/destructive entrepreneurship changes once a state begins offering extraordinarily large economic development incentives. Our empirical approach employs a large database of state incentive awards and a difference-in-differences estimation strategy comparing states that have never awarded these extraordinarily large incentives to those that have awarded them. Because such large incentives are a relatively recent phenomenon, we have data that stretches back to the period before states offered them. As an indication, 2002 was the first year in which a U.S. state (Washington) awarded an incentive valued at more than a billion dollars to a single private firm. With a value in excess of \$32 billion, this single incentive was more than 120,000 times larger than the median incentive award across all states at the time (\$27,000).⁸ In the past two decades, these extraordinarily large incentives went from nonexistent to commonplace in about one-third of U.S. states. As of 2023, more than 30 state-level incentives have topped \$1 billion, and the number of incentives nationwide has skyrocketed from fewer than 100 per year to more than 25,000.⁹

The prior literature on the cost of economic development incentives has generally ignored their public choice and political economy aspects. This gap in the literature has been explicitly noted by authors such as Coyne and Moberg (2014), Jansa and Gray (2017), Jensen and Malesky (2018), and Aobdia, Koester and Petacchi (2021). The expansion of extraordinarily large incentive programs would predictably result in firms devoting additional resources toward securing them; this is true whether or not the firms are successful in their pursuit. While it is difficult to know how many firms waste resources competing unsuccessfully, estimates suggest that at least three compete for every one incentive granted and, further, that firms devoting more resources toward the political process are more likely to be successful (Gabe and Kraybill, 1998; Aobdia, Koester and Petacchi, 2021). Along these lines, Sobel, Wagner and Calcagno (2022) do indeed find that states offering extraordinarily large incentives experience increases both in campaign contributions and in the vote shares of incumbent politicians seeking re-election.

Selective development incentives also create problems beyond just wasteful lobbying and unproductive entrepreneurship. By making certain business activities artificially more profitable than

reductions in taxes or regulations that apply equally to all firms.

⁸The first billion-dollar incentive to a single firm was also more than 3,100 times larger than the mean subsidy award across all states at the time it was awarded (\$1,031,950).

⁹Source: Good Jobs First Subsidy Track database.

others, these incentives distort how resources are used in the economy. This type of government favoritism, often called “crony capitalism,” pushes investment away from the preferences of consumers that would naturally emerge in a free market. As a result, the overall economy becomes less efficient and society’s wellbeing suffers.

3 Data and Empirical Strategy

3.1 Identifying ‘Extraordinarily Large’ Incentives

We combine information on state economic development incentives from Good Job First’s Subsidy Tracker database with a national source of geocoded establishment data from the National Establishment Time Series (NETS) database to explore a causal link between the timing of a state’s first extraordinarily large development incentive and changes in unproductive entrepreneurship, proxied by the number of establishments and employment in the lobbying industry.

The Subsidy Tracker database is, to our knowledge, the most comprehensive source of historical information on development incentives. It includes incentives in state public records, as well as additional incentives compiled by media sources and freedom of information requests. The database consists of nearly 700,000 distinct awards dating as far back as 1975. Major data fields include the total value of individual incentive awards (as reported on public documents), recipient firms, location, funding program(s), and whether an award derives from state, federal, or local funding sources.¹⁰

If a state had to pass legislation to award its first extraordinarily large incentive, it would be easy to identify the year that it did so. In reality, however, identifying the timing of the first such incentive is more difficult. While there is a clear and drastic increase in the magnitude of these incentives within our sample period, most were simply granted under the umbrella of existing state development programs. For example, as Jensen (2019) notes, 238 cities submitted bids during the heated contest to host Amazon’s second headquarters (HQ2). Of the 26 bids that were publicly released, only two required the state to pass funding legislation as part of the application. All other

¹⁰We thank Philip Mattera from Good Jobs First for granting us access to the complete database. A limited number of data fields are publicly available at <https://subsidytracker.goodjobsfirst.org> [accessed: 3/20/2024]. Unfortunately, the Subsidy Tracker database does not provide specific details on the breakdown of the amounts and types of incentives. However, we do see variations in the amounts and recipients that trigger our thresholds across the United States. For instance, International Sematech received \$210,000,000 in New York in 2002, Rhode Island in 2010 gave 38 Studios \$75,000,000, and in Michigan in 2015, Ford Motor Company received \$1.1 billion. All of these packages, while having some detail, are all classified as Megadeals in the Subsidy Tracker.

(publicly released) Amazon bids, the largest of which was valued at \$8.5 billion, were authorized under existing programs even though their magnitude vastly exceeded any prior award offered.

Because many extraordinarily large development incentive offers are hidden from public sight until the deal is consummated, we follow the data-driven algorithm proposed by Sobel, Wagner and Calcagno (2022) to identify the timing of the first in a given state. The basic concept is simple – if a state grants an incentive to a single firm that *substantially* deviates from its own past behavior, then economic agents may view this as a policy shift and respond by engaging in more rent seeking and other unproductive activities. Said differently, the first time a state awards an incentive that is extraordinarily large relative to what it has done in the past, firms and entrepreneurs may reasonably believe that the relative rates of return between productive and unproductive activities have changed and adjust their behavior accordingly.

Following Sobel, Wagner and Calcagno (2022), a state is considered to be “treated” *if* it ever awards an incentive to a single firm that is 3500 times greater than the state’s historical median incentive (as a fraction of state GDP).¹¹ This algorithm captures the first time that a state gives a single incentive that *significantly* deviates from its own past behavior. In contrast, if a state *never* gives a single incentive that crosses the threshold, then the state is considered to be part of the comparison group. Figure 1 illustrates the “treated” and “comparison” states using the data-driven algorithm.

[Figure 1 about here]

Using the full history of the Subsidy Tracker database, 17 states have awarded a single incentive at least 3500 times larger than their historical median award. These states constitute our treatment group. Consider Texas and Connecticut. Texas provided a single incentive to Amazon in 2012 valued at \$269 million, while Connecticut offered a \$400 million incentive to United Technologies Corporation in 2014. At the time these awards were given, Texas’s and Connecticut’s median historical subsidy awards equaled \$40,056 and \$76,250, respectively. Thus, these individual awards were 6,715 times larger than the state’s historical norm in Texas and 5,245 times larger in Connecticut. Therefore, under the data-driven algorithm, both states are classified as “treated” in the year those awards were made.

¹¹State GDP normalizes for differences in the size of state economies. A \$1 billion incentive is a much smaller share of state GDP in Florida than it is in Rhode Island.

Conversely, 33 states have never (to date) awarded a single incentive that was at least 3500 times larger than their historical median award. These states constitute our comparison group. Consider, for instance, Georgia and Iowa. Georgia provided a single incentive valued at \$110 million to NCR in 2009, and Iowa provided an \$84 million incentive to the Iowa Fertilizer Company in 2012. Although these awards may seem large in dollar value, Georgia’s award was “only” 275 times larger than its median historical award at the time (\$400,000), and Iowa’s was “only” 420 times larger than its median award at the time (\$200,000).

Just to be clear, all the comparison group states give economic development incentives. These states are part of the comparison group because their incentive-giving behavior never *substantially* deviated from what they had historically done.¹² The difference-in-differences strategy we employ, outlined in more detail below, will capture how a substantial *change* in incentive-giving behavior affects the *change* in a state’s unproductive entrepreneurship. In the comparison states, those with no significant change in the magnitude of incentive granting, one would not expect to observe a significant shift in the magnitude of unproductive entrepreneurial activity.

Interestingly, there also appears to be a high degree of spatial clustering among the states that have awarded extraordinarily large incentives. For instance, in the south-central region of the U.S., Alabama, Mississippi, Texas, Oklahoma, Kansas, and Missouri all awarded their first extraordinarily large incentive within a three-year window (2009 to 2012). A similar pattern exists for the two other clusters of states, one in the northeast, and one in the west/northwest, that have awarded large development incentives.

Although it is outside the scope of this paper to explore the factors affecting states’ decisions to grant large incentives, the timing and close clustering suggest that yardstick competition or bidding wars play a role. Extraordinarily large development incentives often attract a great deal of favorable press coverage for public officials. The electric vehicle company Rivian’s decision to locate a manufacturing facility outside of Atlanta, Georgia, after receiving billions in subsidies is one such example (Williams, 2022). Once a neighboring state announces a deal, elected officials may feel as though their state must also offer large incentives to remain competitive. While academics have

¹²There is obviously some degree of arbitrariness defining a “significant” or “substantial” deviation from past behavior. Sobel, Wagner and Calcagno (2022) use a range of thresholds to explore the sensitivity of defining treatment status. We use the 3500 times threshold because this falls in the mid-range of thresholds considered by Sobel, Wagner and Calcagno (2022).

voiced concerns about the inefficiencies and inequities in bidding wars for decades (Hanson, 1993), there are no signs of change in these behaviors (Farren and Jean, 2021).

3.2 Measuring Unproductive Entrepreneurship

As we note in Section 2, few studies apart from Dove (2021) and Tuszynski and Stansel (2018) have empirically explored whether there is a correlation between economic development incentives and unproductive entrepreneurship. To our knowledge, this study is the first to use establishment-level data and a quasi-experimental design to estimate a causal effect.

We use individual establishment-level data in the lobbying industry North American Industry Classification System (NAICS 541820) from the National Establishment Time Series (NETS) database to measure unproductive entrepreneurship. This allows us to draw comparisons to Sobel and Garrett (2002) and Sobel (2008), who use statewide (or aggregate) employment and establishments in the lobbying sector as the proxy for unproductive activity. The six-digit NAICS is the most detailed classification for the industry, which includes establishments that provide political consulting, public relations, as well as lobbying services.

NETS was developed by Walls & Associates, LLC, and is a private-sector alternative to the Census Bureau’s restricted-use Longitudinal Business Database. NETS is derived from Dun and Bradstreet’s (D&B) Marketing Information files that track more than 71 million establishments annually (in January) (Walls & Associates, 2019). D&B tracks individual establishments to create data products that can be sold to other business lines, such as marketing research firms. Information is collected from public sources such as court records and secretaries of state, as well as direct contacts by telephone (Barnatchez, Crane and Decker, 2017). Individual establishments are assigned a unique identifier – a DUNS number – so that each can be tracked over time. Until April 2022, the federal government identified individual establishments located in the U.S. by their DUNS number, indicating the number’s reputation for accuracy.¹³

In addition to providing every establishment’s primary 6-digit (NAICS) code, the NETS database includes, among other data points, the establishment’s latitude and longitude, employment count, D&B’s proprietary PAYDEX credit score, whether the establishment has relocated, and the first and last year of active operation. NETS data have been used in numerous studies including, but not limited to, broadband access and establishment growth (Wagner and Lee, 2023), local en-

¹³Since April 2022, the federal government has identified unique establishments using a Unique Entity ID (UEI).

trepreneurship (Neumark, Wall and Zhang, 2011), national monument designation (Walls, Lee and Ashenfarb, 2020), and recent trends in business concentration (Rossi-Hansberg, Sarte and Trachter, 2020).

There are several distinct advantages to measuring unproductive entrepreneurship using lobbying industry data from the NETS database. First, while counts of registered lobbyists are available for some states at the state level, these data are incomplete and mutable as states change their lobbyist registration requirements (Strickland, 2019). Second, there is widespread variation in state reporting requirements, with some states only requiring individuals to register as a lobbyist if they interact directly with elected officials (Louisiana) and others requiring all employees of a lobbyist firm to be disclosed (California) regardless of whether they engage directly with elected officials.¹⁴

For these reasons, industry-level NETS data on lobbyist establishments and employment are more likely than counts of registered lobbyists to provide a consistent measure of the lobbying industry’s (unproductive) activities both within and across states over time.

Additionally, because the NETS data also provide detailed location information for establishments over time, we can leverage it to explore if and how the spatial distribution of lobbying activity changes in a state after the first extraordinarily large incentive is authorized. While we have argued that the first such incentive shifts the relative returns of productive and unproductive activities and, further, that it may lead to more unproductive entrepreneurship, it is also possible that the award causes lobbying activities to become more concentrated in certain areas, such as the state capital. Such an analysis is not possible with aggregated counts of registered lobbyists or with indices, such as Sobel (2008), that capture only a single point in time.

As one might expect, lobbying activity tends to be highly concentrated in a small number of locations in most states. Figure 2 illustrates the counties in each state with the highest share of statewide lobbyist employment averaged over the period from 1997 to 2001 (before New York’s first extraordinarily large incentive was awarded in 2002).

[Figure 2 about here]

In nearly all states, lobbying industry employment is concentrated in the large population centers and the capital county. The county with the largest share of (statewide) lobbyist employment

¹⁴See <https://www.ncsl.org/ethics/lobbyist-registration-requirements> for state information on lobbyist reporting requirements [accessed: 11/18/2023].

is shown in blue, while the capital county is delineated with a thick black border. There are only 18 states in which the capital county has the highest (or maximum) concentration of lobbyist employment, and across all states, capital counties are home to an average of only 24% of a state’s total employment in that industry. Instead, the geographic focal point for lobbying activity is most likely to be the most populated county in a state’s most populated metropolitan region.

Nationwide, a very large number of counties, 2,586 (or 82.3%), are home to less than 0.5% of the statewide lobbyist industry. These counties are the lightly shaded ones in Figure 2. In contrast, fewer than 7% of the counties nationwide (or 210) are home to 5% or more of their state’s lobbyist employment (shaded more darkly). The median state has four counties with lobbyist employment shares of at least 5% of the statewide total, and these counties account for an average of almost 87% of total lobbyist employment. Very few industries are so highly geographically concentrated as lobbying.

3.3 Empirical Specification

We employ a difference-in-differences identification strategy to estimate the causal effect of the timing of a state’s first extraordinarily large incentive on establishment employment and concentration in the lobbying industry. The general form of the difference-in-differences model is given by:

$$Y_{ist} = \alpha + \beta D_{ist} + X_{st}\theta + RT_t\phi + E_{is}\nu + \epsilon_{ist} \quad (1)$$

where Y_{ist} is the outcome of interest for establishment i , located in state s , at time t . X_{st} is a vector of observable, time-varying state characteristics that may be correlated with the trend growth in lobbyist employment or concentration. RT_t denotes the relative time (or time to treatment) fixed effects that will adjust for shocks common to all establishments. Establishment-level fixed effects (E_i) will adjust for time-invariant factors at the establishment level, which would include such factors as (time-invariant) local regulations affecting the industry.

D_{ist} is the difference-in-differences variable of interest; it equals unity if establishment i is located within a treated state (or county) after the year in which the first large incentive was awarded, and equals zero otherwise. β is the difference-in-differences parameter of interest. It captures the average change in lobbyist employment or concentration in states that have awarded extraordinarily large incentives compared to the change in lobbyist employment or concentration

in states that have never awarded such an incentive.

Individual states awarded large incentives at different points in time. Given the limitations of the NETS data, which ends in 2019, individual establishments are included in the regressions if they are observed at least one year before and after treatment. We also initially limit the sample to establishments that never relocated during the sample period (more than 98% of all establishments) so that we are comparing same-location establishments in treated locations to same-location establishments in comparison locations. We explore new lobbying establishment births and relocations when we explore capital county concentration in Section 4.5.

We utilize the “stacked regression” difference-in-differences approach proposed by Cengiz et al. (2019) to address the staggered adoption. This estimator is intuitive and simple to implement because estimation is done by Ordinary Least Squares. More importantly, as simulations by Baker, Larcker and Wang (2022) and Gardner (2021) show, average treatment effect estimates from the stacked estimator are numerically equivalent to the estimators proposed by Callaway and Sant’Anna (2021), Sun and Sant’Anna (2021), and Gardner (2021) if there are heterogeneous treatment effects across groups or time.

The underlying idea behind the stacked regression strategy is the formation of a comparison group for each treatment group cohort, so that “early” treated states never become part of the comparison group. In other words, the stacked estimator adjusts the sample so that establishments in treated states are always compared to “clean comparisons” (establishments that are never treated) by collapsing the regression to a single treatment period for all treated establishments (relative time = 0). If there are heterogeneous treatment effects across states or time and one fails to adjust for the staggered awarding of large incentives, then the estimate of the average treatment effect would be biased because establishments treated “early” in the sample end up serving as part of the comparison group for establishments treated “later” in the sample, sometimes called the “forbidden comparison” (Roth et al., 2023).

The stacked estimator sample is formed in two steps. First, a “dataset” is constructed for every treatment group cohort in the sample (or distinct treatment year). Next, each cohort’s dataset is combined into a single large dataset. Consider 2009, when Kansas and Utah awarded their first extraordinarily large incentives. This cohort’s dataset consists of all lobbyist establishments in Kansas and Utah (the treated states) plus all lobbyist establishments in the 33 untreated states

(never treated). Treatment occurs at time 0 (or 2009), and relative time -1 and +1 refer to 2008 and 2010, respectively. This process is repeated for each distinct year in which a state (or states) authorized the first large incentive, and the final dataset is formed by “stacking” (row concatenating) the individual cohort datasets.

The stacked data structure achieves two objectives. First, it avoids the “forbidden comparison” because treated units are never part of the comparison group. Second, it guarantees that every treated cohort is treated in the same relative time period (period 0).¹⁵ For each treatment group cohort (with treatment at time 0), the sample includes observations ranging from -5 to +5. Thus, in the case of the 2009 cohort noted above, observations range from 2004 (-5) to 2014 (+5).

The identifying assumption in a difference-in-differences regression, which is untestable, is that trend lobbyist employment or concentration in the comparison states is a valid counterfactual for what would have occurred in lobbyist employment or concentration in the treated states in the absence of awarding an extraordinarily large incentive.

Recent work by Strickland (2019) suggests that institutional features play a role in explaining differences in lobbying activity within and across states. We include several time-varying covariates that could shed light on trend differences in lobbyist activity. First, we include the share of each state’s senate and house seats that are up for election in a given year. This will adjust for differences in lobbying intensity that occur during election cycles (Horgos and Zimmermann, 2009). Next, we include an indicator variable if the state legislature is controlled by Democrats (yes = 1), and an indicator variable if the state has binding term limits (yes = 1). Strickland (2019) finds that one-party dominance and the presence of legislative term limits are correlated with fewer registered lobbyists, all else being equal. Finally, based on recent work by Strickland (2019), Ban and You (2019), and Campos and Giovannoni (2017), we include several variables related to campaign finance reporting that may influence the intensity of lobbying activity. These variables include: the maximum candidate-to-candidate contributions (in dollars) permitted, the minimum contribution amounts from a single donor (in dollars) that trigger name disclosure, the maximum amount (in dollars) of public funding available to candidates for office, and an indicator variable if corporations must also file a separate expense report (yes = 1).

¹⁵As Cengiz et al. (2019) note, the stacked regression estimated by OLS must be fully saturated with (group by cohort) and (time by cohort) fixed effects to recover an unbiased estimate of the average treatment effect.

Descriptive statistics and source information are provided in Table 1. Figures in Table 1 are for 2008, which is roughly the midpoint of our empirical sample.

[Table 1 about here]

The average number of employees in the lobbying industry, NAICS 541820, is 5,904 in the treated states and 2,526 in the comparison states. The treated states are, on average, more populous than the comparison states, which explains the level difference in industry sizes. On a per capita basis, lobbyist employment is very similar across treated and comparison states. Moreover, on a per establishment basis, which is how the data enter the regressions, lobbyist employment averages (in 2008) 4.9 workers in treated states and 4.7 in comparison states.

Although lobbying employment as a share of total industry employment tends to be highly concentrated in a handful of counties, the overall industry remains a very small share of total economic activity. Specifically, lobbying employment and establishments account for, on average, less than 0.25% of either total employment or establishments in capital counties.

4 Empirical Results

4.1 Establishment Results

We begin by exploring how the timing of a state’s first extraordinarily large incentive affects establishment-level employment. Table 2 presents the difference-in-differences estimates. The dependent variable is the log of the number of employees, and regressions have been estimated with establishment fixed effects, relative time (or time-to-treatment) fixed effects, and a constant term that is not reported. Standard errors, shown in parentheses, are clustered at the state dimension.

Estimates for all establishments located in treated states are shown in column 1. Given the spatial variation in industry concentration (see Figure 2), we also show results for establishments located in capital counties (column 2) and establishments located in the county with the highest lobbyist concentration (column 3). We also compare employment in every state’s least concentrated third of counties (column 4), middle third of concentrated counties (column 5), and highest third of concentrated counties (column 6).¹⁶

¹⁶The distribution and maximum counties for each state were based on the five-year average from 1997 to 2001, which pre-dates the timing of the first extraordinarily large incentive.

Recall that lobbyist establishments that relocated during our sample period (fewer than 2%) are excluded from these estimates. This allows us to compare establishments that operated before and after the first large incentive in treated states to establishments that operated in the same areas (capital counties, etc.) in the comparison states.

[Table 2 about here]

Since the dependent variable is the log of establishment employment, the average percentage change in establishment-level employment after the first extraordinarily large incentive was awarded (relative to establishments in the comparison states) is given by $100 * (e^{\beta} - 1)$, where β is the parameter of interest (Giles et al., 2011). For small coefficient estimates, such as those in Table 2, the percentage change is closely approximated with $100 * \beta$.

In general, the point estimates in Table 2 show that establishment-level employment in the lobbying industry increases significantly after the first large incentive. In column 1, for instance, the estimates indicate that (on average) employment increases by 3.6% per establishment compared to the group states that have never (to date) awarded an extraordinarily large incentive.

The results in columns 2 through 5 reveal salient information about where the lobbying industry growth occurs. For example, we find no evidence that industry employment significantly increases in areas of the treated states with either a low (bottom 33%) or mid (33 to 66%) concentration of lobbyist jobs, as shown in columns 4 and 5. The gains are all concentrated in counties that have the highest concentration of lobbyist jobs (the top third), with the average establishment located in these counties experiencing employment gains of 3.8% (column 6).

As to what is driving the effect in highly concentrated counties, that seems to be employment gains in capital counties. Relative to the comparison states, we find that establishments operating in the state’s capital county before the first extraordinarily large incentive experience employment increases averaging 4.5% in the first five years after the extraordinarily large was awarded (column 2). This is consistent with our assertion that a state’s first extraordinarily large economic development incentive may be perceived by firms and entrepreneurs as a shift in incentive-giving behavior and an indication that unproductive activities (such as lobbying) are more lucrative than productive ones. Firms and entrepreneurs then reallocate resources accordingly.

The point estimates in Table 2 have a causal interpretation if the (untestable) parallel trends

assumption is valid. We re-estimated the establishment-level regressions using an event study specification to explore any pre-treatment differences between the treated and comparison states. The event study regressions include the same covariates and (group x cohort) and (year x cohort) fixed effects. Relative time period -1 is the omitted indicator.

[Figure 3 about here]

The event study plots do not indicate significant pre-treatment differences for any of the regressions, suggesting that establishment-level employment in the comparison states is a valid counterfactual for how establishment-level employment would have evolved in treated states in the absence of the extraordinarily large award.

4.2 Varying Treatment Thresholds

States are considered treated in the preceeding section when they award a single subsidy to one firm that is 3500 times larger than their historical median award. While this may *seem* like a very large threshold that applies only to a small number of states, 17 states have provided incentive awards meeting this criterion. In this section, we explore the robustness of our findings by varying the treatment threshold from 2000 to 5000 times larger than the state’s own historical median award. In 21 states, at least one incentive package was more than 2000 times bigger than their typical incentive amount. Even more dramatically, 13 states have offered incentives that were over 5000 times larger than usual.

Table 3 reports the difference-in-differences parameter of interest when we vary the treatment threshold. These regressions include the same covariates reported in Table 2 but we have omitted them for brevity. The column headings in Table 3 also correspond exactly to the same samples reported in each column in Table 2.

[Table 3 about here]

We find very similar results in terms of economic magnitude and statistical significance when we vary the treatment thresholds. This indicates that our estimates are not an artifact of a single threshold. Additionally, as the threshold to trigger treatment increases from 2000 to 5000, the number of “treated” states varies from a high of 21 to a low of 13 because fewer states meet the more stringent criterion. This is evidence that the expanded concentration we find in lobbying

activity after the first extraordinarily large incentive is a general response and is not being driven by the behavior of a small number of states.

All of the empirical results that follow from this point in the paper are based on the 3500 threshold trigger used in Table 2.

4.3 Business and Professional Organizations

While the main results in Section 4.1 focus on employment in the lobbying sector, economic development incentives may also induce rent-seeking in various business associations representing their members. This section of the paper expands the focus to establishments in industries that represent business and professional associations. Business associations (NAICS 813910) focus on promoting the interests of businesses and industries as a whole, with the most common examples being Chambers of Commerce and industry trade groups. In contrast, professional associations (NAICS 813920) focus on the interests of individual professionals and maintaining professional standards. Common examples include the American Medical Association and state bar associations. Estimates for business associations are reported in Table 4, while results for professional associations are reported in Table 5.

[Tables 4 and 5 about here]

We find limited evidence that business and professional organization employment expands after a state awards its first extraordinarily large incentive relative to these same organizations in comparison states. The one exception is column 2 in Table 4, where we find that establishments representing business associations in capital counties experience employment increases of 2% relative to comparison establishments. The point estimates are positive in each specification for both business and professional organizations, but they are too imprecisely estimated to conclude that they differ from zero.

There are several potential reasons why there is limited evidence that business and professional organization employment expands after the state's first extraordinarily large incentive. First, business and professional organizations represent their members on wide range of issues apart from just economic development incentives. For example, licensure requirements and other trade restrictions may be a higher priority than a tax break. Second, business and professional organizations may simply have a preference to hire lobbyists rather than expand their own employment. If there is

a risk that business organizations or firms may be viewed negatively by the public for directly engaging with government officials, then hiring a lobbyist to advance one’s agenda provides at least some level of concealment over direct engagement.

4.4 Robustness Check Excluding Neighboring States

The baseline results presented in Table 2 include all untreated states in the comparison group cohorts. This is problematic for our empirical strategy if one state’s decision to award an extraordinarily large subsidy incentivizes (or disincentivizes) other states to offer their own because such an effect would bias our estimates. Put differently, if a state awards its first extraordinarily large incentive and nearby states believe they must expand their incentive-giving behavior to “keep up,” lobbying activity in nearby states could also expand (regardless of whether a large incentive is actually offered). This scenario seems plausible given the spatial clustering of states that have awarded these large incentives, as shown in Figure 2.

To rule out bias from neighboring states, we re-estimate the baseline regressions and exclude any state from the comparison group that borders a treated state. As an illustration, Indiana never awarded an extraordinarily large incentive during the sample, so it is included in the comparison group in Table 2. In this section, we exclude Indiana from the empirical analysis altogether because it shares a border with Michigan, which awarded an extraordinarily large incentive in 2010. Once we remove all border states from the comparison group, the total pool of comparison states shrinks from 33 to 12.

[Table 6 about here]

Table 6 shows that the results using this alternative comparison group are consistent with those using the full comparison group (Table 2). That is, we find that establishment-level lobbying employment expands in a state after it awards an initial extraordinarily large incentive. These alternative results also corroborate the fact that lobbying employment increases most in capital counties.

4.5 Industry Concentration Results

The results from Sections 4.1 and 4.4 indicate that establishments in capital counties experience the largest employment gains. In this section, we explore whether the lobbying industry becomes

more concentrated in capital counties relative to overall private-sector economic activity. In these regressions, we also include lobbyist employment, new establishments, and establishments that relocated to the capital county.

We include the same covariates in the concentration regressions as in the previous section. Because we are now focusing only on the capital county in each state, the (group x cohort) fixed effects are (state x cohort) fixed effects in the stacked regressions.

[Table 7 about here]

Column 1 in Table 7 shows estimates measuring lobbying concentration as the share of total private-sector capital county employment. Similarly, the estimates in column 2 measure concentration of lobbying industry establishments as the share of the total number of (private-sector) establishments located in the capital county. Standard errors are clustered at the state dimension.

Although the lobbying industry is a small share of overall economic activity in capital counties, the estimates in Table 7 show that employment concentration significantly increases after the awarding of the state’s first extraordinarily large incentive. We find, for instance, that the concentration of lobbyist employment increases by 5.8% (column 1), while the concentration of lobbyist establishments, including new establishments and movers, remains unchanged relative to the comparison states (column 2). This implies that existing lobbyist establishments in the capital counties capture the “gains” in unproductive activity that result from states awarding their first extraordinarily large incentive.

[Figure 4 about here]

We also estimate event study versions of the regressions in Table 7. The plots include the covariates and omit the relative time -1 indicator variable.

Consistent with the establishment-level plots in Figure 3, these findings show no evidence that lobbying concentration differed between treated and comparison states in the pre-treatment period. This suggests that the comparison states are a valid counterfactual for potential outcomes in the treated states.

5 Conclusion

State economic development incentives have been used to attract and retain private firms for decades. However, in about one-third of U.S. states, the size of the incentives given to single firms recently has exploded into the billions of dollars. While the prior literature finds little to no evidence that these extraordinarily large development incentives improve overall economic outcomes, there is clear evidence from Sobel, Wagner and Calcagno (2022) of the gains to the politicians who grant them – gains that come in the form of higher campaign contributions and incumbent reelection vote shares. We build on the findings of Sobel, Wagner and Calcagno (2022) and Dove (2021) to examine if, in addition, the awarding of these extraordinarily large incentives is associated with an environment that generates more unproductive entrepreneurship consistent with Baumol (1990), Sobel (2008), and Trubnikov (2021).

Using geocoded establishment data for lobbyist firms from 1997 to 2019 as a measure of unproductive entrepreneurial activity and a staggered difference-in-differences approach, we test whether lobbyist establishments and concentration grow in states once they begin offering extraordinarily large incentives. We follow the method in Sobel, Wagner and Calcagno (2022) to identify the years for which the 17 states that began offering these extraordinarily large incentives did so within our sample period and thus are considered treated, leaving 33 comparison states.

Our estimates suggest that employment increases on average by 3.6% in lobbyist establishments in states that begin offering extraordinarily large incentives. When examining the geographic breakdown, we find, as expected, that these gains are largely driven by employment in state capital counties. We find that - in the five years after the award of such a large incentive - employment increases on average 4.5 percent more in treated states’ capital counties than it does in the capital counties of comparison states.

In addition, we find that lobbying industry concentration significantly increases after a state begins offering extraordinarily large incentives. Specifically, our estimates suggest that the concentration of lobbyist employment increases by 5.8% and, further, that the number of lobbyist establishments remains unchanged. In other words, industry “gains” appear to be completely captured by existing establishments. Besides finding evidence that existing lobbyist employment increases for establishments in the capital county, we find evidence that the industry as a whole

becomes a significantly larger share of overall private-sector economic activity following the offer of an extraordinarily large incentive.

While the evidence that state economic development incentives improve broad economic outcomes is scant at best, the incentives certainly benefit the firms that receive them, their shareholders, and the politicians who grant them. Indeed, even though the prior literature argues these incentive programs are negative sum owing to their unintended consequences and secondary effects, few studies have attempted to quantify these consequences and effects. As our examination shows, subsequent to states' offering extraordinarily large incentives, there is an expansion in unproductive activities, specifically lobbyist employment, and concentration. Rather than accomplishing their stated goal of promoting productive economic activities, extraordinarily large incentives lead to the allocation of resources toward a path of unproductive activity.

Finally, since state and local governments offer many different types of incentive programs, it would be valuable to investigate whether certain incentives are more likely than others to encourage wasteful business activities or attempts to capture government subsidies without creating tangible economic benefits.

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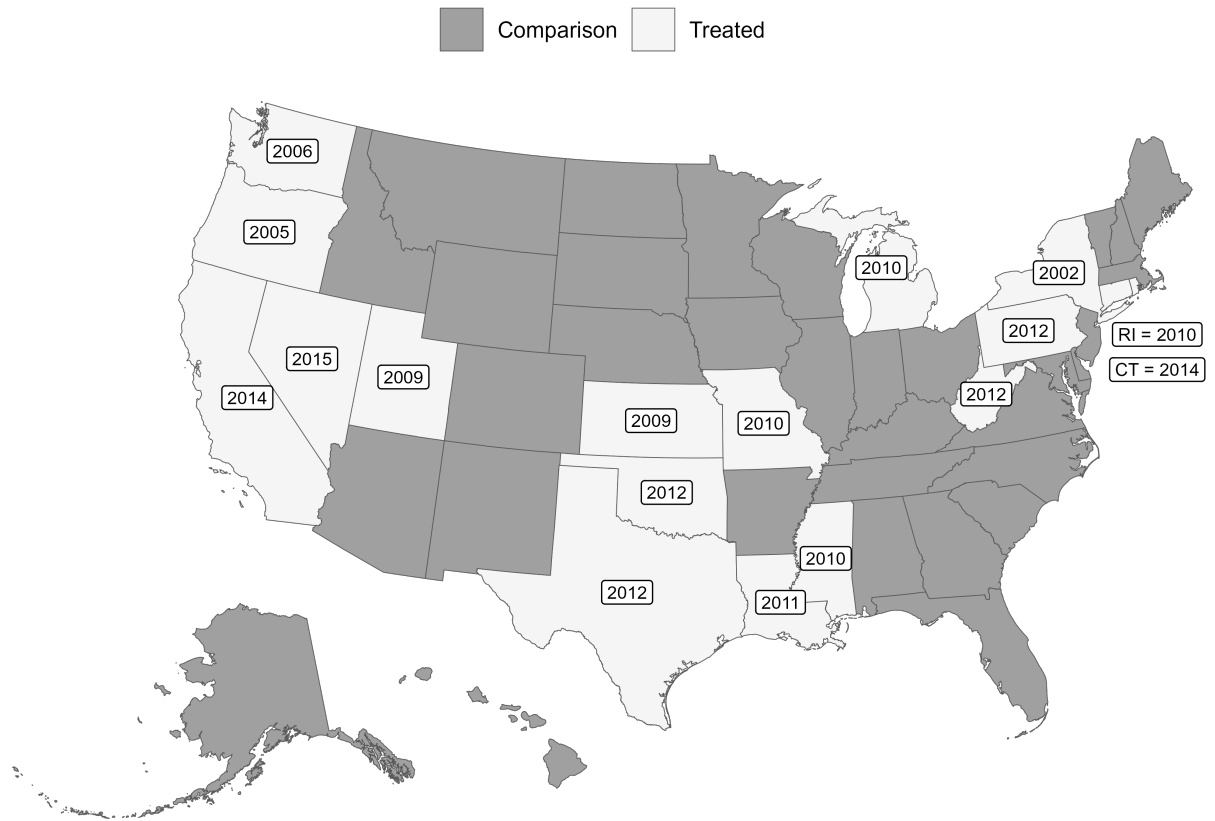
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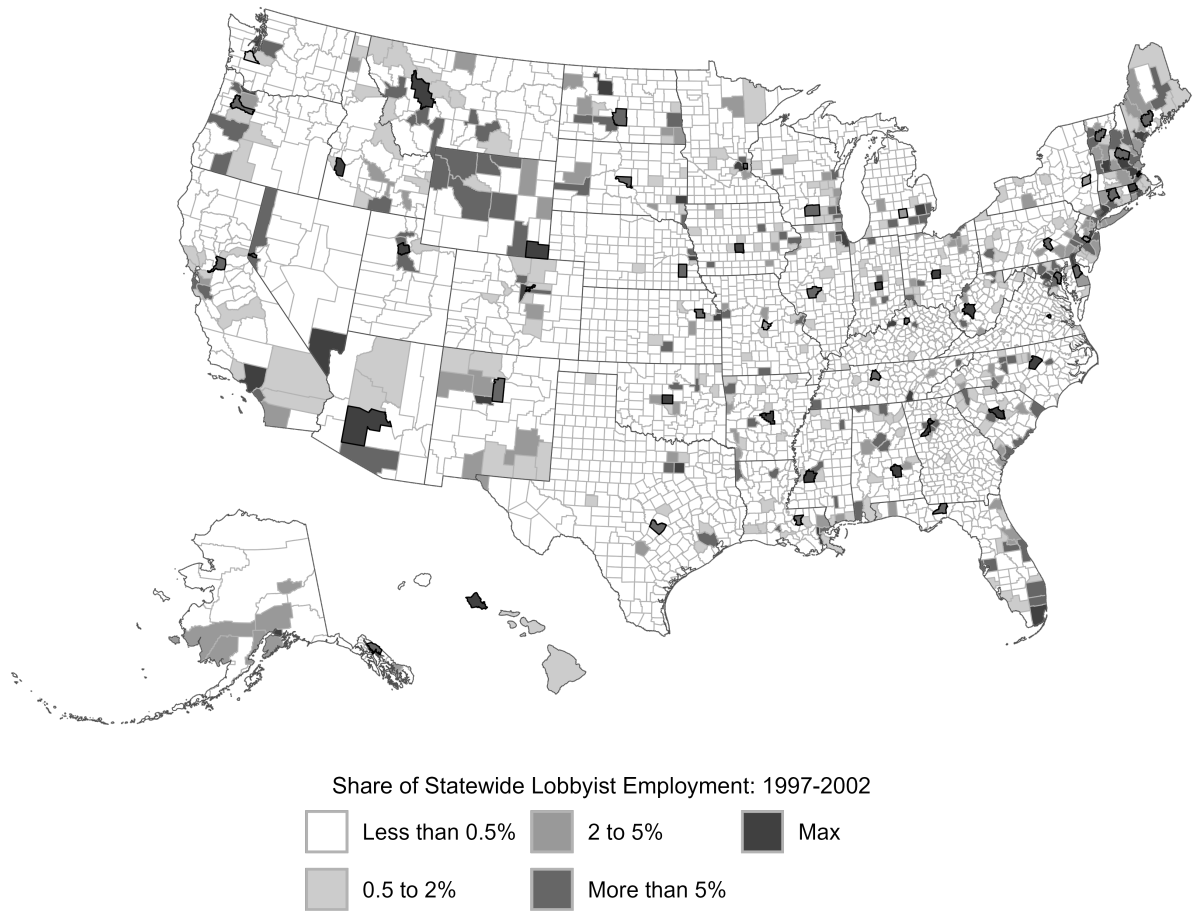
Figure 1: Treated and Comparison States



Source: Sobel, Wagner, and Calcagno (2022).

Note: This figure shows the year in which states awarded their first extraordinarily large economic development incentive to a single firm. As noted in the text, an extraordinarily large incentive is defined as the state's first incentive that is awarded to a single private-sector establishment that is 3500 times (or more) greater than the state's past historical median award value.

Figure 2: Distribution of State Lobbyist Employment



Note: This figure shows the concentration of state lobbyist employment, defined by NAICS 541820, at the county dimension. Each county's share of total state lobbyist employment is the average from 1997 to 2001, which pre-dates the first extraordinarily large state economic development incentive. Capital city counties are delineated by a thick black border. The county in each state with the maximum concentration of lobbyist employment is shown in blue.

Table 1: Descriptive Statistics and Data Sources

Variable	treatment mean	treatment std	comparison mean	comparison std
	mean	std dev	mean	std dev
statewide lobbyist employment	5904.529	8488.219	2526.455	3092.977
statewide lobbyist establishments	843.647	1226.282	423.909	528.656
capitall county lobbyist employment	629.294	837.318	657.212	898.395
capitall county lobbyist establishments	88.706	83.119	89.333	101.532
Democrat legislative control	0.206	0.367	0.303	0.413
binding legislative term limits	0.294	0.470	0.273	0.452
senate seats up for election	0.532	0.329	0.598	0.340
house seats up for election	0.882	0.332	0.833	0.368
candidate to candidate contribution limit	7264.706	17104.712	4052.364	8544.101
mandatory donor disclosure trigger	66.882	50.893	90.879	98.662
maximum public candidate funding	0.000	0.000	2438.545	14008.377
corporations file independent reports	0.529	0.514	0.515	0.508

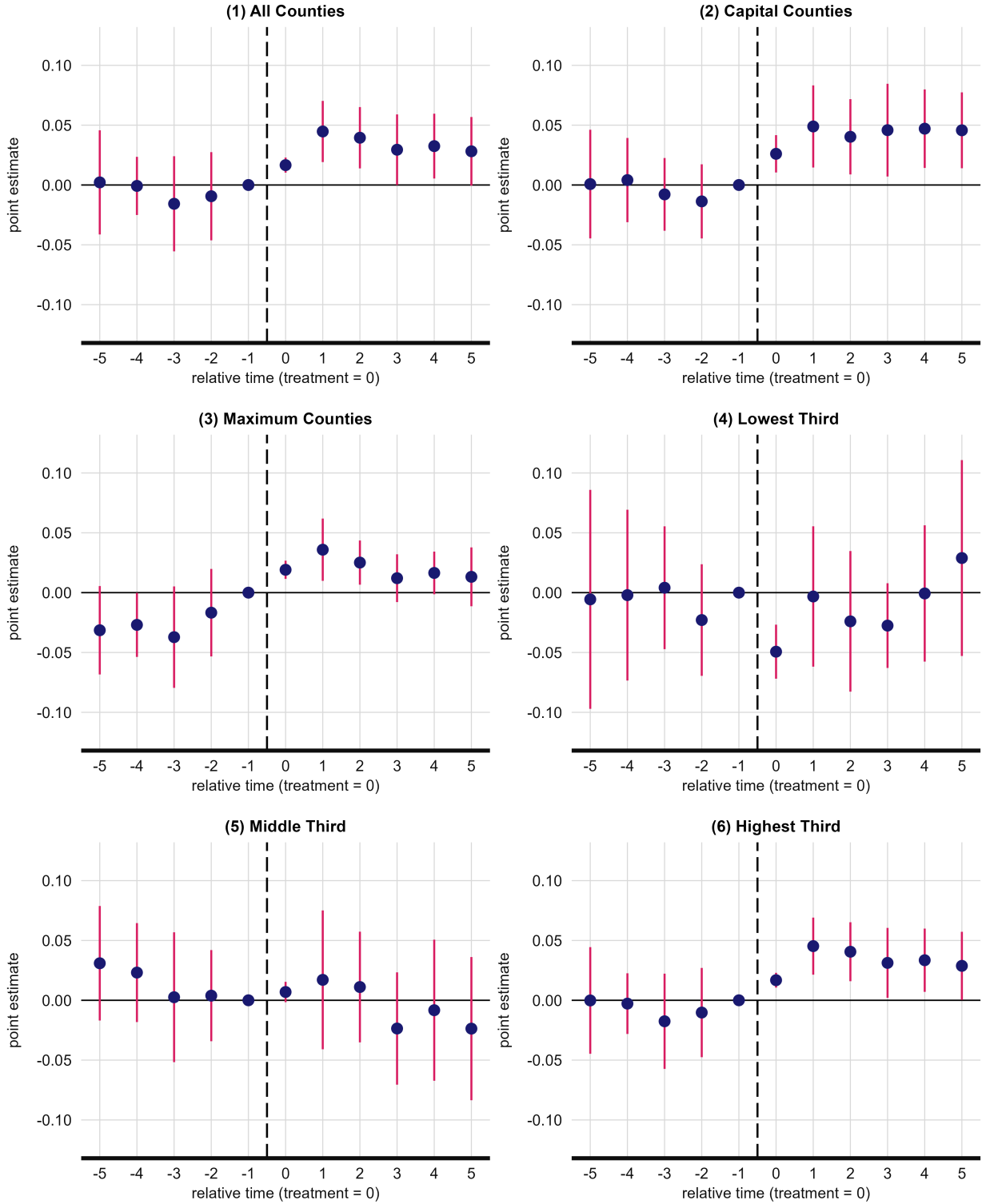
This table descriptive statistics for the 17 treated and 33 comparison states shown in Figure 1. Due to the staggered timing of when states awarded their first extraordinarily large economic development incentive, descriptive statistics show in this table are for 2008 (roughly the sample midpoint). Lobbyist employment and establishment counts are from the National Establishment Time Series database. Democrat control of the legislature (yes = 1) and the fraction of house and senate seats up for election are from the Correlates of State Policy Project database. The year that state legislative term limits became binding is from the National Conference of State Legislature. All of the remaining variables are from the Campaign Finance Institute.

Table 2: Effect of the Initial Extraordinary Large Incentive
on Lobbyist Establishment Employment

	<i>Dependent variable:</i> log(employment)					
	(1) OLS	(2) OLS	(3) OLS	(4) OLS	(5) OLS	(6) OLS
D	0.036*** (0.014)	0.045*** (0.011)	0.042*** (0.015)	-0.013 (0.019)	-0.011 (0.024)	0.038*** (0.014)
Democrat legislative control	-0.030*** (0.005)	-0.037*** (0.006)	-0.023*** (0.007)	0.004 (0.013)	-0.021* (0.011)	-0.032*** (0.006)
binding legislative term limits	-0.036*** (0.012)	-0.029 (0.023)	-0.032*** (0.009)	-0.025*** (0.003)	-0.029* (0.017)	-0.036*** (0.012)
senate seats up for election	0.001 (0.008)	-0.021** (0.011)	-0.016** (0.008)	-0.005 (0.022)	0.026* (0.015)	-0.002 (0.008)
house seats up for election	-0.008 (0.007)	0.005 (0.008)	0.004 (0.006)	-0.011 (0.016)	-0.034** (0.016)	-0.005 (0.006)
candidate to candidate contribution limit	0.004 (0.003)	0.003 (0.003)	0.004 (0.003)	0.011** (0.006)	0.019*** (0.006)	0.004 (0.003)
mandatory donor disclosure trigger	-1.488* (0.778)	-1.995*** (0.285)	-0.705 (1.060)	-3.042*** (1.151)	-1.034 (1.197)	-1.468* (0.839)
maximum public candidate funding	-0.005 (0.004)	-0.005 (0.006)	-0.007 (0.005)	-0.006*** (0.002)	0.005*** (0.001)	-0.006 (0.005)
corporations file independent reports	3.946 (14.346)	10.906 (9.752)	3.451 (15.507)	41.112 (27.325)	-30.236 (23.454)	5.206 (14.467)
N	1,325,087	245,152	387,408	40,162	92,702	1,200,208
Year x cohort FE	yes	yes	yes	yes	yes	yes
Establishment x cohort FE	yes	yes	yes	yes	yes	yes
Adj. R^2	0.917	0.919	0.926	0.862	0.892	0.918
Treated establishments	12,397	1,284	4,112	258	469	11,711
Comparison establishments	27,361	4,882	7,648	1006	2043	24,527
Sample	All Counties	Capital Counties	Max Counties	Lowest Third	Middle Third	Highest Third

This table presents difference-in- differences estimates of the effect of the timing of a state's first extraordinarily large economic development incentive on individual lobbyist establishment employment (NAICS 541820). D is the difference-in-differences parameter of interest. Standard errors are clustered at the state dimension. *** denotes significance at the 1 percent level, ** at the 5 percent level, and * at the 10 percent level. Regressions were estimated with a constant term, (establishment x cohort) fixed effects, and (year x cohort size) fixed effects that are not reported. Column (1) includes lobbyist establishments in every county within a state, column (2) includes only the counties where the capital city is located, column (3) includes only establishments located in the county with the highest concentration of lobbyist employment in the state, column (4) includes establishments in each state's bottom third in the distribution of lobbyist employment, column (5) includes establishments in each state's middle third in the distribution of lobbyist employment, and column (6) includes establishments in each state's highest third in the distribution of lobbyist employment.

Figure 3: Event Study Plots



Note: This figure shows the event study plots for the main regression results in Table 2. Numbers in parentheses correspond to the column numbers in Table 2. Error bars show the 95% confidence intervals.

Table 3: Robustness Check: Varying “Extraordinary Large Incentive” Threshold

	<i>Dependent variable:</i> log(employment)					
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: 2000 x Historical Median Award						
<i>21 states met or exceeded this threshold</i>						
D	0.033*** (0.013)	0.045*** (0.010)	0.038*** (0.014)	-0.041* (0.023)	-0.004 (0.022)	0.035*** (0.013)
Panel B: 2500 x Historical Median Award						
<i>19 states met or exceeded this threshold</i>						
D	0.036*** (0.013)	0.047*** (0.011)	0.041*** (0.014)	-0.009 (0.018)	-0.009 (0.023)	0.038*** (0.013)
Panel C: 3000 x Historical Median Award						
<i>18 states met or exceeded this threshold</i>						
D	0.036*** (0.014)	0.043*** (0.011)	0.041*** (0.015)	-0.017 (0.018)	-0.011 (0.024)	0.038*** (0.014)
Panel D: 4000 x Historical Median Award						
<i>17 states met or exceeded this threshold</i>						
D	0.035** (0.014)	0.043*** (0.012)	0.040*** (0.015)	-0.014 (0.019)	-0.014 (0.024)	0.037*** (0.014)
Panel E: 4500 x Historical Median Award						
<i>16 states met or exceeded this threshold</i>						
D	0.034** (0.014)	0.045*** (0.012)	0.039** (0.016)	-0.023 (0.021)	-0.020 (0.024)	0.036** (0.014)
Panel F: 5000 x Historical Median Award						
<i>13 states met or exceeded this threshold</i>						
D	0.036** (0.015)	0.049*** (0.013)	0.040** (0.016)	-0.031 (0.021)	-0.021 (0.024)	0.038*** (0.015)

This table presents difference-in- differences estimates of the effect of the timing of a state’s first extraordinarily large economic development incentive on individual lobbyist establishment employment (NAICS 541820). D is the difference-in-differences parameter of interest. Standard errors are clustered at the state dimension. *** denotes significance at the 1 percent level, ** at the 5 percent level, and * at the 10 percent level. Regressions were estimated with a constant term, (establishment x cohort) fixed effects, and (year x cohort size) fixed effects that are not reported. Column (1) includes lobbyist establishments in every county within a state, column (2) includes only the counties where the capital city is located, column (3) includes only establishments located in the county with the highest concentration of lobbyist employment in the state, column (4) includes establishments in each state’s bottom third in the distribution of lobbyist employment, column (5) includes establishments in each state’s middle third in the distribution of lobbyist employment, and column (6) includes establishments in each state’s highest third in the distribution of lobbyist employment.

Table 4: Business Association Establishment Employment

	<i>Dependent variable:</i> log(employment)					
	(1) OLS	(2) OLS	(3) OLS	(4) OLS	(5) OLS	(6) OLS
D	0.021 (0.019)	0.021** (0.011)	0.024 (0.025)	0.016 (0.015)	0.020 (0.022)	0.022 (0.019)
Democrat legislative control	-0.035*** (0.007)	-0.019 (0.014)	-0.033*** (0.008)	-0.010 (0.017)	-0.045*** (0.008)	-0.035*** (0.007)
binding legislative term limits	-0.004 (0.031)	-0.012 (0.011)	-0.014 (0.013)	0.026 (0.047)	0.007 (0.073)	-0.008 (0.022)
senate seats up for election	-0.006 (0.007)	-0.021 (0.013)	-0.016 (0.013)	-0.030*** (0.011)	-0.004 (0.008)	-0.004 (0.007)
house seats up for election	-0.009 (0.006)	0.004 (0.009)	0.003 (0.010)	-0.007 (0.007)	-0.013* (0.007)	-0.008 (0.006)
candidate to candidate contribution limit	-0.001 (0.002)	0.005 (0.003)	-0.004*** (0.001)	0.019*** (0.005)	0.001 (0.003)	-0.002 (0.002)
mandatory donor disclosure trigger	-2.433*** (0.752)	-3.598*** (0.658)	-1.440 (1.109)	-3.913*** (1.416)	-2.564** (1.021)	-2.279*** (0.739)
maximum public candidate funding	-0.001 (0.001)	-0.002 (0.002)	0.000 (0.001)	0.000 (0.002)	-0.002 (0.002)	0.000 (0.001)
corporations file independent reports	13.876 (10.918)	22.315* (12.784)	13.284 (14.113)	24.175 (21.193)	13.914 (9.989)	13.335 (11.315)
N	2,513,113	371,284	547,747	178,358	422,495	1,913,543
Year x cohort FE	yes	yes	yes	yes	yes	yes
Establishment x cohort FE	yes	yes	yes	yes	yes	yes
Adj. R^2	0.872	0.895	0.892	0.789	0.826	0.881
Treated establishments	20,067	2,094	3,839	1,051	2,800	16,231
Comparison establishments	43,072	6,144	9,433	3,092	7,109	32,886
Sample	All Counties	Capital Counties	Max Counties	Lowest Third	Middle Third	Highest Third

This table presents difference-in- differences estimates of the effect of the timing of a state's first extraordinarily large economic development incentive on individual business associations employment (NAICS 813910). D is the difference-in-differences parameter of interest. Standard errors are clustered at the state dimension. *** denotes significance at the 1 percent level, ** at the 5 percent level, and * at the 10 percent level. Regressions were estimated with a constant term, (establishment x cohort) fixed effects, and (year x cohort size) fixed effects that are not reported. Column (1) includes lobbyist establishments in every county within a state, column (2) includes only the counties where the capital city is located, column (3) includes only establishments located in the county with the highest concentration of lobbyist employment in the state, column (4) includes establishments in each state's bottom third in the distribution of lobbyist employment, column (5) includes establishments in each state's middle third in the distribution of lobbyist employment, and column (6) includes establishments in each state's highest third in the distribution of lobbyist employment.

Table 5: Professional Organizations Establishment Employment

	<i>Dependent variable:</i>					
	log(employment)					
	(1)	(2)	(3)	(4)	(5)	(6)
	OLS	OLS	OLS	OLS	OLS	OLS
D	0.019 (0.018)	0.020 (0.012)	0.020 (0.020)	0.042 (0.029)	0.036 (0.024)	0.017 (0.018)
Democrat legislative control	-0.041*** (0.008)	-0.025 (0.019)	-0.039*** (0.012)	-0.053* (0.027)	-0.038*** (0.013)	-0.040*** (0.008)
binding legislative term limits	-0.099*** (0.019)	-0.030 (0.036)	-0.099*** (0.030)	-0.610* (0.354)	-0.098*** (0.029)	-0.100*** (0.020)
senate seats up for election	0.001 (0.010)	0.004 (0.012)	-0.007 (0.016)	-0.024 (0.026)	0.003 (0.017)	0.003 (0.010)
house seats up for election	-0.013 (0.010)	-0.012 (0.010)	-0.001 (0.016)	0.012 (0.017)	-0.021 (0.017)	-0.013 (0.010)
candidate to candidate contribution limit	0.001 (0.003)	0.001 (0.004)	0.000 (0.002)	0.007 (0.017)	0.003 (0.009)	0.000 (0.003)
mandatory donor disclosure trigger	-1.110 (0.897)	-2.282*** (0.512)	-0.376 (1.042)	-2.220* (1.146)	-0.882 (1.159)	-1.072 (0.914)
maximum public candidate funding	0.001 (0.002)	0.003* (0.002)	0.002 (0.002)	-0.002 (0.003)	-0.010*** (0.003)	0.004** (0.002)
corporations file independent reports	-30.514 (30.083)	14.555 (10.221)	-20.479 (33.683)	-30.333 (26.920)	-32.961 (35.000)	-30.347 (30.322)
N	1,502,828	266,250	369,126	55,022	158,527	1,292,572
Year x cohort FE	yes	yes	yes	yes	yes	yes
Establishment x cohort FE	yes	yes	yes	yes	yes	yes
Adj. R^2	0.897	0.900	0.910	0.835	0.878	0.899
Treated establishments	12,425	1,473	2,957	356	959	11,122
Comparison establishments	28,789	4,716	6,863	1,296	3,270	24,311
Sample	All Counties	Capital Counties	Max Counties	Lowest Third	Middle Third	Highest Third

This table presents difference-in- differences estimates of the effect of the timing of a state's first extraordinarily large economic development incentive on individual professional business organization establishment employment (NAICS 813920). D is the difference-in-differences parameter of interest. Standard errors are clustered at the state dimension. *** denotes significance at the 1 percent level, ** at the 5 percent level, and * at the 10 percent level. Regressions were estimated with a constant term, (establishment x cohort) fixed effects, and (year x cohort size) fixed effects that are not reported. Column (1) includes lobbyist establishments in every county within a state, column (2) includes only the counties where the capital city is located, column (3) includes only establishments located in the county with the highest concentration of lobbyist employment in the state, column (4) includes establishments in each state's bottom third in the distribution of lobbyist employment, column (5) includes establishments in each state's middle third in the distribution of lobbyist employment, and column (6) includes establishments in each state's highest third in the distribution of lobbyist employment.

Table 6: Robustness Check: Omitting Border States from the Comparison Group

	<i>Dependent variable:</i> log(employment)					
	(1) OLS	(2) OLS	(3) OLS	(4) OLS	(5) OLS	(6) OLS
D	0.032** (0.014)	0.044*** (0.011)	0.034** (0.014)	-0.004 (0.020)	-0.006 (0.024)	0.033** (0.014)
Democrat legislative control	-0.035*** (0.006)	-0.033*** (0.007)	-0.040*** (0.011)	0.024 (0.020)	-0.017* (0.010)	-0.039*** (0.007)
binding legislative term limits	-0.018 (0.014)	0.036 (0.032)	-0.058*** (0.018)	-0.015 (0.021)	-0.006** (0.003)	-0.019 (0.015)
senate seats up for election	-0.004 (0.009)	-0.013 (0.010)	-0.001 (0.007)	-0.024 (0.052)	-0.056*** (0.021)	-0.001 (0.008)
house seats up for election	-0.001 (0.007)	0.003 (0.011)	-0.007 (0.008)	0.020 (0.041)	0.036*** (0.013)	-0.003 (0.007)
candidate to candidate contribution limit	0.005 (0.014)	0.002 (0.012)	0.002 (0.016)	0.007 (0.016)	0.014 (0.013)	0.005 (0.014)
mandatory donor disclosure trigger	-0.440 (0.742)	-1.991*** (0.571)	0.124 (0.866)	-0.573 (0.915)	-5.241** (2.608)	-0.416 (0.751)
maximum public candidate funding	-0.003 (0.005)	-0.005 (0.006)	-0.005 (0.005)	-0.007** (0.003)	0.005*** (0.001)	-0.004 (0.006)
corporations file independent reports	-34.731 (41.564)	-4.582 (37.868)	-32.088 (53.319)	41.678 (50.426)	-27.838 (36.099)	-35.129 (42.616)
N	685,454	106,324	193,996	15,749	43,176	629,578
Year x cohort FE	yes	yes	yes	yes	yes	yes
Establishment x cohort FE	yes	yes	yes	yes	yes	yes
Adj. R^2	0.914	0.919	0.922	0.831	0.881	0.916
Treated establishments	12,397	1,284	4,112	258	469	11,711
Comparison establishments	13,279	1,919	3,498	348	940	12,061
Sample	All Counties	Capital Counties	Max Counties	Lowest Third	Middle Third	Highest Third

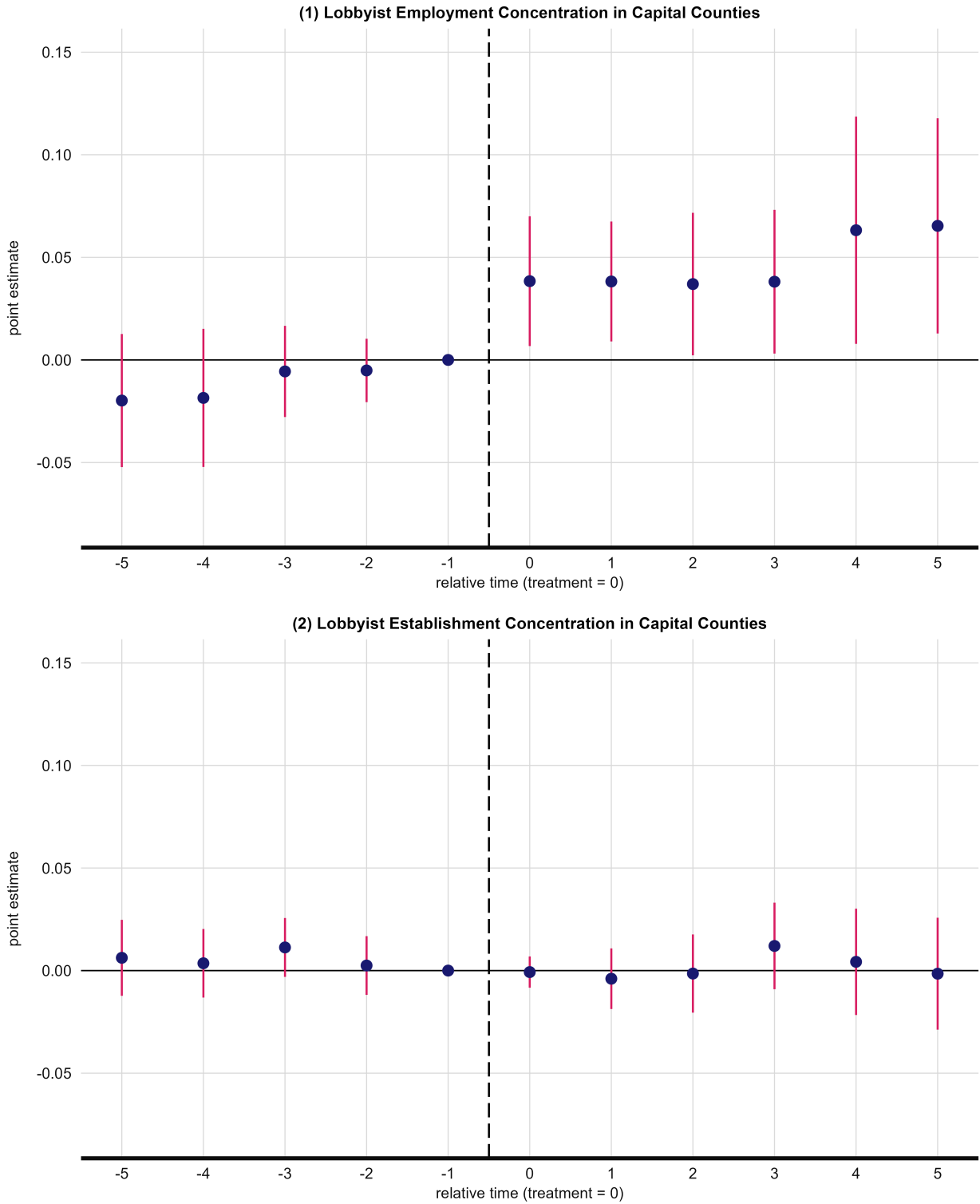
This table presents difference-in-differences estimates of the effect of the timing of a state's first extraordinarily large economic development incentive on individual lobbyist establishment employment (NAICS 541820). States that border a treated state have been omitted from the comparison group. D is the difference-in-differences parameter of interest. Standard errors are clustered at the state dimension. *** denotes significance at the 1 percent level, ** at the 5 percent level, and * at the 10 percent level. Regressions were estimated with a constant term, (establishment x cohort) fixed effects, and (year x cohort size) fixed effects that are not reported. Column (1) includes lobbyist establishments in every county within a state, column (2) includes only the counties where the capital city is located, column (3) includes only establishments located in the county with the highest concentration of lobbyist employment in the state, column (4) includes establishments in each state's bottom third in the distribution of lobbyist employment, column (5) includes establishments in each state's middle third in the distribution of lobbyist employment, and column (6) includes establishments in each state's highest third in the distribution of lobbyist employment.

Table 7: Effect of the Initial Extraordinary Large Incentive
on Capital County Lobbyist Concentration

	<i>Dependent variable:</i>	
	employment share	establishment share
	(1) OLS	(2) OLS
D	0.058*** (0.021)	-0.003 (0.010)
Democrat legislative control	0.031** (0.015)	0.014 (0.009)
binding legislative term limits	0.005 (0.008)	-0.047 (0.031)
senate seats up for election	-0.022 (0.016)	-0.002 (0.008)
house seats up for election	0.026 (0.016)	0.021** (0.008)
candidate to candidate contribution limit	-0.010 (0.007)	-0.013*** (0.004)
mandatory donor disclosure trigger	0.067 (0.524)	1.094* (0.630)
maximum public candidate funding	-0.005** (0.002)	-0.002 (0.002)
corporations file independent reports	34.619 (21.352)	-5.370 (8.828)
N	3,420	3,420
Year x cohort FE	yes	yes
State x cohort FE	yes	yes
Adj. R^2	0.605	0.850
Sample	Employment Concentration	Establishment Concentration

This table presents difference-in-differences estimates of the effect of the timing of a state's first extraordinarily large economic development incentive on the concentration of lobbyist activity in the state's capital county. D is the difference-in-differences parameter of interest. Standard errors are clustered at the state dimension. *** denotes significance at the 1 percent level, ** at the 5 percent level, and * at the 10 percent level. Regressions were estimated with a constant term, (establishment x cohort) fixed effects, and (year X cohort size) fixed effects that are not reported. Column (1) shows the effect on lobbyist employment as a share of total private-sector employment, while column (2) shows the effect on the share of private-sector establishments in the lobbyist sector.

Figure 4: Event Study Plots



Note: This figure shows the event study plots for the regression results in Table 7. Numbers in parentheses correspond to the column numbers in Table 7. Error bars show the 95% confidence intervals.